A Counterexample to the Liminf Random-Sidon Weakening

fa_banach_001 candidate packet

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Status

Status: **counterexample_likely_valid**.

This packet concerns Problem 8 in Section 6 of Aihua Fan, Herve Queffelec, and Martine Queffelec, *Old and new results on the Furstenberg sets*, arXiv:2303.06850v1.

Source Question

The source paper studies random sets

$$R = \{n \in \mathbb{N} : \xi_n = 1\}, \qquad \mathbb{E}\xi_n = \delta_n,$$

where the selectors are independent and (δ_n) is decreasing. It sets $m_N = \sum_{n \le N} \delta_n$. Theorem 5.12 proves that if $m_N = O(\log N)$, then R is almost surely Sidon.

Problem 8 asks whether the theorem remains true under only

$$\liminf_{N\to\infty} \frac{m_N}{\log N} < \infty.$$

The problem appears on page 30 of the source PDF:

6. Questions on S and T

Problem 1. *Theorem 2.1 and Theorem 2.2 can be stated and proved in the same way for* $S(q_1, q_2)$. *But efforts are needed for* $S(q_1, q_2, q_3)$.

Problem 2. *Is* S *a rigid set ?*

Problem 3. *Is a Sidon set always rigid ?*

Problem 4. *Is a Bohr-dense set always uniformly recurrent ?*

Problem 5. *Is there a Bohr-dense Sidon set ? Equivalently, a non Bohr-closed Sidon set?*

Problem 6. *Is the Furstenberg set* S $\frac{4}{3}$*-Rider (even* $\frac{4}{3}$*-Sidon) ? More generally, is* $S(q_1, \ldots, q_s)$ $\frac{2s}{s+1}$*-Rider (even* $\frac{2s}{s+1}$*-Sidon) ?*

Problem 7. *Is* T *a* $\frac{4}{3}$*-Rider set ?*

Problem 8. *A simple argument shows that, in Theorem 5.12, we can replace* $m_N = O(\log N)$ *by* $m_{N_j} = O(\log N_j)$ *where* (N_j) *is an increasing sequence of integers such that* $N_{j+1} = O(N_j)$. *But we do not know if the assumption* $\liminf_{N\to\infty} m_N/\log N < \infty$ *is enough.*

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References

[1] P. Arnoux, *Sturmian sequences.* Substitutions in dynamics, arithmetics and combinatorics, 143–198, Lecture Notes in Math., 1794, Springer, Berlin, 2002.

[2] C. Badea, S. Grivaux, *Kazhdan constants, continuous probability measures with large Fourier coefficients and rigidity sequences,* Comment. Math. Helv. 95 (2020), no. 1, 99-127.

[3] J. Barral and A. H. Fan *Covering numbers of different points in Dvoretzky covering,* Bull. Sci. Math., vol. 129, no. 4, (2005), 275-317.

[4] A. Bérczes, A. Dujella, L. Hajdu, *Some Diophantine properties of the sequence of S–units.* J. Number Theory 138 (2014), 48–68.

[5] V. Bergelson, A. del Junco, M. Lemanczyk, J. Rosenblatt, *Rigidity and non-recurrence along sequences.* Ergodic Theory Dynam. Systems 34 (2014), 1464–1502.

Proof Intuition

The condition $m_N = O(\log N)$ controls the random counting function at every scale. A bounded liminf controls it only at a sparse sequence of scales. The counterexample exploits this gap: first keep the selector means constant for a long enough plateau that the expected count at the endpoint is much larger than $\log N$; then drop the means to a much smaller level and wait until the denominator $\log N$ catches up. This makes $m_N/\log N$ small infinitely often but large infinitely often. Concentration transfers the large deterministic expectations to the actual random set, and the resulting counting growth is incompatible with Sidonicity.

Counterexample

Theorem 1. *There is a nonincreasing sequence* $(\delta_n) \subset (0, 1]$ *such that, with*

$$m_N = \sum_{n \le N} \delta_n,$$

one has

$$\liminf_{N\to\infty} \frac{m_N}{\log N} < \infty,$$

but the independent selector set

$$R = \{n \in \mathbb{N} : \xi_n = 1\}, \qquad \mathbb{P}(\xi_n = 1) = \delta_n,$$

is almost surely not Sidon. Consequently the proposed liminf-only weakening in Problem 8 is false.

Proof. We use natural logarithms. Choose an integer N_0 so large that $\log N_0 \geq 16$. We construct two increasing integer sequences (H_j) and (N_j), and define (δ_n) in blocks.

Start with $\delta_n = 1/N_0$ for $1 \leq n \leq N_0$. Suppose N_{j-1} has been chosen. Put

$$H_j = \lceil j\, N_{j-1} \log N_{j-1} \rceil$$

and set

$$\delta_n = \frac{1}{N_{j-1}} \qquad (N_{j-1} < n \leq H_j).$$

Next choose $N_j > H_j$ so large that

$$m_{H_j} + 1 \leq \log N_j$$

and also, for the next block,

$$\log((j+1)N_j \log N_j + 2) \leq 2 \log N_j.$$

This is possible because N_j is free to be arbitrarily large. Finally set

$$\delta_n = \frac{1}{N_j} \qquad (H_j < n \leq N_j).$$

The construction covers all positive integers. Since N_j is increasing, the sequence (δ_n) is nonincreasing.

At the low endpoints,

$$m_{N_j} = m_{H_j} + \frac{N_j - H_j}{N_j} \leq m_{H_j} + 1 \leq \log N_j,$$

so

$$\liminf_{N\to\infty} \frac{m_N}{\log N} \leq 1.$$

At the high endpoints,

$$m_{H_j} \geq \frac{H_j - N_{j-1}}{N_{j-1}} \geq j \log N_{j-1} - 1.$$

For $j \geq 2$, the auxiliary condition imposed when N_{j-1} was chosen gives $\log H_j \leq 2 \log N_{j-1}$. Hence

$$\frac{m_{H_j}}{\log H_j} \geq \frac{j \log N_{j-1} - 1}{2 \log N_{j-1}} \longrightarrow \infty.$$

Let $X_N = |R \cap [1, N]|$. For each j, X_{H_j} is a sum of independent Bernoulli variables with mean m_{H_j}. Chernoff's lower-tail inequality gives

$$\mathbb{P}\left(X_{H_j} < \frac{1}{2} m_{H_j}\right) \leq \exp(-m_{H_j}/8).$$

Since $m_{H_j} \geq j \log N_0 - 1$ and $\log N_0 \geq 16$, the right-hand side is summable in j. By Borel–Cantelli, almost surely

$$X_{H_j} \geq \frac{1}{2} m_{H_j}$$

for all sufficiently large j. Therefore

$$\limsup_{N \to \infty} \frac{|R \cap [1, N]|}{\log N} \geq \limsup_{j \to \infty} \frac{X_{H_j}}{\log H_j} = \infty$$

almost surely.

It remains only to recall the standard Sidon mesh obstruction. If $\Lambda \subset \mathbb{N}$ is Sidon, Pisier's criterion, quoted in the source paper's subsection on quasi-independent sets and Sidon sets, gives a constant $\eta > 0$ such that every finite $A \subset \Lambda$ contains a quasi-independent subset $B \subset A$ with $|B| \geq \eta |A|$. If $A = \Lambda \cap [1, N]$, then all subset sums of B are distinct and lie between 0 and $|B|N$. Thus

$$2^{|B|} \leq |B|N + 1,$$

which implies $|B| \leq C \log N$, and hence

$$|\Lambda \cap [1, N]| \leq C_\Lambda \log N$$

for all N. The random set constructed above violates this estimate almost surely, so it is almost surely not Sidon. □

Verification Notes

The proof is deterministic except for one standard Chernoff–Borel–Cantelli step. The block construction gives both required deterministic facts:

$$\liminf_N m_N / \log N \leq 1, \qquad \limsup_j m_{H_j} / \log H_j = \infty.$$

The probabilistic step then transfers the second fact to the random counting function on the selected endpoints.

The crop was generated by

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conda run --no-capture-output -n sandbox python code/make_open_problem_crop.py.
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No computational proof search was used.

Duplicate and novelty checks were bounded as follows. I searched the cheap run indexes for arXiv:2303.06850, "liminf $m_N / \log N$", "Bourgain random condition", "Kahane Katznelson", and random Sidon phrases; no prior packet or attempt was found. Web searches on 2026-07-03 for exact and close phrases such as "liminf m_N log N Sidon random set" and the paper title found the source paper but no later exact resolution. Novelty confidence is therefore *likely valid / needs human review*, not human-verified.

Human Review Recommendation

Check first that Problem 8 is intended as a replacement hypothesis for the almost-sure Sidon conclusion of Theorem 5.12. Under that reading, the counterexample is direct: the selector means satisfy the proposed liminf hypothesis, while the random set almost surely violates the necessary Sidon counting bound.

References

[1] Aihua Fan, Herve Queffelec, and Martine Queffelec. *Old and new results on the Furstenberg sets*. arXiv:2303.06850v1, 2023.